

Project Documentation: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

1. Introduction

- **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- **Team Members & Roles:**
 - **Gabrian John Peter:** Lead Developer / ServiceNow System Administrator (Responsible for Flow Designer configuration, Service Catalog item creation, group routing rules, and UAT test case execution).

2. Project Overview

- **Purpose:** The purpose of this project is to automate the creation and assignment of Catalog Tasks (`sc_task`) for "Standard Laptop" requests within ServiceNow. By leveraging Flow Designer, the workflow triggers immediately upon request approval, routing tasks to the Hardware group without manual intervention to reduce processing speed and eliminate handoff bottlenecks.
- **Features:**
 - Automated trigger execution on Requested Item (`sc_req_item`) approval.
 - Auto-generation of structured Catalog Tasks linked to the parent RITM.
 - Direct task assignment to the **Hardware** assignment group for staging and fulfillment.
 - Audit logging and visual workflow execution tracking.

3. Architecture

- **Frontend / Presentation Layer:** ServiceNow Service Portal / Service Catalog interface built using HTML5/AngularJS components where end users submit request forms.
- **Backend / Application Layer:** ServiceNow **Flow Designer** orchestration engine executing workflow triggers, record creation scripts, and server-side business logic.
- **Database / Data Model:** ServiceNow relational database tables including `sc_req_item` (Requested Item), `sc_task` (Catalog Task), and `sys_user_group` (Assignment Groups).

4. Setup Instructions

- **Prerequisites:**
 - ServiceNow Instance (Tokyo / Utah / Vancouver / Washington DC / Xanadu).
 - Administrator (`admin`) or Flow Designer (`flow_designer`) privileges.
- **Installation & Configuration:**

1. Import the project Update Set into the target ServiceNow instance via **System Update Sets > Retrieved Update Sets**.
2. Preview and commit the Update Set to deploy the **Standard Laptop Task Flow**.
3. Ensure the **Hardware** group (`sys_user_group`) is created and populated with active technicians.
4. Navigate to **Flow Designer**, open **Standard Laptop Task Flow**, and click **Activate**.

5. Folder Structure / Layout

- **Service Catalog Engine:** Service Catalog > Catalog Definitions > Maintain Items (Standard Laptop Request)
- **Process Automation:** Process Automation > Flow Designer (Standard Laptop Task Flow)
- **Task Management:** Service Desk > My Work / Catalog Tasks (`sc_task`)

6. Running the Application

1. Log in to the ServiceNow Service Portal as an end user.
2. Search for **Standard Laptop** in the Hardware Catalog and click **Order Now**.
3. Log in as an approving manager and approve the generated Request Item (`sc_req_item`).
4. Verify in **Flow Designer > Execution History** that the flow triggered and created an `sc_task` assigned to the **Hardware** group.

7. API & Trigger Documentation

- **Trigger Event:** Record Created / Updated on Table `sc_req_item`.
 - **Conditions:** Approval == Approved AND Item == Standard Laptop.
- **Flow Action:** Create Catalog Task
 - **Inputs:** Parent RITM = Triggering Record, Short Description = Configure Standard Laptop, Assignment Group = Hardware.

8. Authentication & Access Control

- Role-Based Access Control (RBAC) configured via ServiceNow Access Control Lists (ACLs).
- Workflow tasks run under the System Context (`system`) to ensure secure execution regardless of user session permissions.

9. User Interface

- **Service Catalog Entry:** Clean, user-friendly request form with configurable laptop options.
- **Flow Designer Execution Canvas:** Visual drag-and-drop workflow layout showing real-time execution states (Completed / In Progress / Waiting).

10. Testing

- **Strategy:** End-to-end User Acceptance Testing (UAT) across functional, security, exception handling, and performance dimensions.
- **Results:** 78 total test cases executed — **78 Passed / 0 Failed**.
- **Defect Resolution:** 77 pre-UAT defects resolved (37 Fixed, 20 By Design, 8 Won't Fix, 6 External, 4 Duplicates, 2 Skipped, 1 Not Reproduced).

11. Screenshots or Demo

- **Project Workspace & Configuration Demo:** [SkillWallet Project Workspace](<https://myskillwallet.ai/dashboard/skillwallet/module/servicenow-system-administrator-nm-eng-6a69e3438beabdd402737035/group-projects/6a96be23a12a0aa741756191/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-with-Flow-Designer-6ab62ae4fa5e5219dc38c865?tab=3>)[cite: 5, 6]

12. Known Issues

- None remaining for core flow execution; all critical (Severity 1) and high (Severity 2) issues resolved prior to submission.

13. Future Enhancements

- Integration with external vendor REST APIs (e.g., Dell/HP procurement portals) for real-time shipment updates.
- Automated Slack / Microsoft Teams webhooks to notify technicians when a new hardware task is created.